

Semiparametric Inference and Sensitivity Analysis for Causal Mediation in Linear Models with Unspecified Error Distributions

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Abstract

Causal mediation analysis distinguishes pathways through which a treatment affects an outcome. In linear models, ordinary least squares may leave information in the error distribution unused. We develop a semiparametric framework for joint inference in linear mediation models with treatment–mediator interaction and unspecified error distributions. A stacked estimating system integrates regression-error scores and propagates uncertainty from both regressions and the baseline-covariate distribution through natural-effect decompositions. Under the stated model and regularity conditions, we characterize the natural-effect influence functions and establish an OLS-relative asymptotic variance and local-power ordering. We further develop density-score estimation and joint inference for residual-correlation sensitivity analysis of unmeasured mediator–outcome confounding. Reduced-form regressions and joint uncertainty propagation for their coefficients, residual variances and cross-covariances yield pointwise sensitivity curves and variance-explained contours under explicit working assumptions. Simulations show reduced root mean squared error across several non-Gaussian settings and markedly higher power for a weak indirect effect under asymmetric-mixture errors. The power advantage over ordinary least squares and interaction-adapted Huber comparators persists after independent null calibration. Evaluations also examine interval coverage, numerical success, full sensitivity curves and variance misspecification. An analysis of the randomized JOBS II behavioral intervention illustrates how natural-effect estimates, uncertainty and confounding sensitivity can be interpreted together. Complete estimating equations, a deterministic multi-start algorithm and reproducible numerical output accompany the analysis.

Keywords: causal mediation; natural effects; sensitivity analysis; semiparametric estimation; unspecified error distributions

1 Introduction

Natural direct and indirect effects provide counterfactual descriptions of how an exposure may operate through a mediator (Robins and Greenland, 1992; Pearl, 2001; Imai et al., 2010a; VanderWeele, 2015). In linear models these effects reduce to familiar functions of coefficients from a mediator regression and an outcome regression. Ordinary least squares (OLS) offers a convenient way to estimate these coefficients without specifying an error distribution. A further question is whether estimating the shape of that distribution can improve precision when the regression model is correctly specified. Skewed and multimodal errors make this question relevant even within a simple linear mediation analysis.

With treatment–mediator interaction, the total effect admits two equally valid natural-effect decompositions (Valeri and VanderWeele, 2013; VanderWeele, 2015). The pure natural direct effect (PNDE) combines with the total natural indirect effect (TNIE), while the total natural direct effect (TNDE) combines with the pure natural indirect effect (PNIE). Each component remains a single population quantity. Distinguishing these decompositions is important both substantively and computationally, because all four components depend on coefficients from linked regressions.

Existing semiparametric mediation methods provide several routes to natural-effect inference. Tchetgen Tchetgen and Shpitser (2012) derived efficiency bounds and multiply robust estimators, Chan et al. (2016) developed nonparametric estimation through balancing weights, and Farbmacher et al. (2022) combined efficient scores with sample splitting and machine learning for flexible covariate adjustment. We address a complementary question: how estimated error-density scores can improve natural-effect inference within linked linear models. The regression model in Section 2.1 defines the corresponding efficiency benchmark and permits an explicit characterization of the information gain over OLS.

A separate literature addresses distributional robustness in regression-based mediation. Yuan and MacKinnon (2014) used median regression, while Alfons et al. (2022) implemented robust regression and bootstrap procedures in `robmed`; Alfons and Schley (2025) review the different notions of robustness. Wang et al. (2025) developed a Huber-loss approach balancing robustness and efficiency. Parametric structural equation models also accommodate skewness through specified non-Gaussian families (Asparouhov and Muthén, 2016). Here the target is a mean-based natural effect, and the error density is estimated without selecting a paramet-

ric family. The focus is on extracting error-shape information for precision, complementing methods designed for resistance to contamination or high-leverage observations.

Building on the regression scores of Kim (2023), this article connects error-density information to joint mediation inference, precision and detection of weak effects. Three contributions are central. First, a joint estimating system for the two regressions and the baseline-covariate mean yields natural-effect influence functions and propagates uncertainty through both decompositions. Second, an explicit asymptotic variance gap identifies when the feasible estimator improves on OLS and links that gain to local power. Third, observation-level score equations, deterministic multi-start computation and joint covariance estimation provide a reproducible procedure. Its evaluation covers precision, coverage and numerical success, including matched-data comparisons with OLS, the quasi-Bayesian and percentile-bootstrap procedures in the R package `mediation`, and natural effect models in the R package `medflex`. Independently calibrated power comparisons also include interaction-adapted Huber estimators. Section 2.1 states the statistical and causal assumptions.

A further extension develops density-score estimation and joint inference for the residual-correlation sensitivity functional of Imai et al. (2010a). Related work establishes semiparametric sensitivity formulations (Tchetgen Tchetgen and Shpitser, 2012) and, more recently, debiased-machine-learning procedures with R^2 contours and robustness summaries (Liu and McCann, 2026). Within this literature, our reduced-form construction accommodates unspecified marginal error densities and propagates the complete covariance of regression coefficients, residual variances, cross-covariances and baseline means through the sensitivity functional. It provides pointwise sensitivity curves, residual-variance contours and zero-boundary inference, together with a full-curve Monte Carlo assessment. The analysis combines efficient marginal regression scores under the working model with a joint influence-function representation for sensitivity inference.

Section 2 develops the estimation, inference and OLS-relative precision and local-power results. Sections 3 and 4 evaluate the procedure and its confounding-sensitivity extension, followed by the JOBS II application and discussion. Online Resource 1 (Supplementary Material) follows the same sequence and supplies proofs, implementation details and complete numerical results.

2 Semiparametric estimation and inference

2.1 Model and natural-effect preliminaries

Let the observed data be independent copies of $O = (\mathbf{X}, T, M, Y)$, where $T \in \{0, 1\}$ is treatment, M is a continuous mediator, Y is a continuous outcome, and $\mathbf{X}$ contains pretreatment covariates. We posit

$$M = \alpha_2 + \beta_2 T + \boldsymbol{\xi}_2^\top \mathbf{X} + \epsilon_M, \quad (1)$$

$$Y = \alpha_3 + \beta_3 T + \gamma M + \eta T M + \boldsymbol{\xi}_3^\top \mathbf{X} + \epsilon_Y. \quad (2)$$

For estimation we assume $\epsilon_M \perp\!\!\!\perp (T, \mathbf{X})$, $\epsilon_Y \perp\!\!\!\perp (T, M, \mathbf{X})$, zero error means, finite positive variances, and otherwise unknown continuously differentiable error densities. Thus the errors are homoscedastic with freely varying non-Gaussian shapes. These restrictions define the statistical model used for the efficiency calculation, while the causal interpretation follows from the identification assumptions below.

Write $M(t)$ for the potential mediator under treatment t and $Y(t, m)$ for the potential outcome under treatment t and mediator value m . We use consistency, positivity, and sequential ignorability:

Assumption 1 (Sequential ignorability). *For $t, t' \in \{0, 1\}$ and relevant $(m, \mathbf{x})$,*

$$\{Y(t', m), M(t)\} \perp\!\!\!\perp T \mid \mathbf{X}, \quad Y(t', m) \perp\!\!\!\perp M(t) \mid T = t, \mathbf{X} = \mathbf{x}.$$

The second condition rules out unmeasured mediator–outcome confounding after conditioning on treatment and baseline covariates. It remains substantive even when treatment is randomized.

Define

$$\delta(t) = \mathbb{E}\{Y(t, M(1)) - Y(t, M(0))\},$$

$$\zeta(t) = \mathbb{E}\{Y(1, M(t)) - Y(0, M(t))\},$$

$$\tau = \mathbb{E}\{Y(1, M(1)) - Y(0, M(0))\}.$$

The conventional labels are pure natural indirect effect (PNIE), total natural indirect effect (TNIE), pure natural direct effect (PNDE), total natural direct effect (TNDE), and total effect

106 (TE):

$$\text{PNIE} = \delta(0), \quad \text{TNIE} = \delta(1), \quad \text{PNDE} = \zeta(0), \quad \text{TNDE} = \zeta(1), \quad \text{TE} = \tau.$$

107 Let $\boldsymbol{\mu}_X = \mathbb{E}(\mathbf{X})$, $\mu_0 = \alpha_2 + \boldsymbol{\xi}_2^\top \boldsymbol{\mu}_X$, and $\mu_1 = \mu_0 + \beta_2$. Under Assumption 1 and models
 108 (1)–(2), standard mediation identification gives (Imai et al., 2010a; Valeri and VanderWeele,
 109 2013; VanderWeele, 2015)

$$\begin{aligned} \text{PNIE} &= \beta_2 \gamma, & \text{TNIE} &= \beta_2(\gamma + \eta), \\ \text{PNDE} &= \beta_3 + \eta \mu_0, & \text{TNDE} &= \beta_3 + \eta \mu_1, \\ \text{TE} &= \beta_3 + \beta_2 \gamma + \eta \mu_1. \end{aligned} \tag{3}$$

110 Consequently,

$$\text{TE} = \text{PNDE} + \text{TNIE} = \text{TNDE} + \text{PNIE}. \tag{4}$$

111 When $\eta = 0$, PNIE equals TNIE and PNDE equals TNDE, so the two decompositions coincide.
 112 Section S1 of Online Resource 1 gives the identification calculation.

113 2.2 Error-density scores and feasible estimation

114 The regression score derived by Kim (2023, Theorem 1) supplies the building block for the joint
 115 mediation construction. Consider a generic regression

$$R = m(\mathbf{Z}; \boldsymbol{\beta}) + \epsilon, \quad \mathbb{E}(\epsilon) = 0, \quad \text{Var}(\epsilon) = v, \quad \epsilon \perp\!\!\!\perp \mathbf{Z}, \tag{5}$$

116 where f is the unknown error density and $\dot{m}_\beta(\mathbf{Z}; \boldsymbol{\beta}) = \partial m(\mathbf{Z}; \boldsymbol{\beta}) / \partial \boldsymbol{\beta}$. Put

$$q(\epsilon) = \epsilon^2 - v - \frac{\mu_3}{v} \epsilon, \quad \mu_3 = \mathbb{E}(\epsilon^3), \quad Q_2 = \mathbb{E}\{q(\epsilon)^2\}. \tag{6}$$

117 We assume f is positive and continuously differentiable on the relevant support, $\mathbb{E}(\epsilon^4) < \infty$, $0 <$
 118 $Q_2 < \infty$, the location Fisher information is finite, the regression design has full-rank information,
 119 and the population efficient-score equation has a locally unique root. These conditions ensure
 120 nondegenerate score information and local identification of the regression parameters. The

121 efficient score of Kim (2023) is

$$S_{\beta}^{\text{eff}} = -\frac{f'(\epsilon)}{f(\epsilon)} [\dot{m}_{\beta}(\mathbf{Z}; \beta) - \mathbb{E}\{\dot{m}_{\beta}(\mathbf{Z}; \beta)\}] + \left\{ \frac{\epsilon}{v} - \frac{\mu_3 q(\epsilon)}{v Q_2} \right\} \mathbb{E}\{\dot{m}_{\beta}(\mathbf{Z}; \beta)\}, \quad (7)$$

$$S_v^{\text{eff}} = \frac{q(\epsilon)}{Q_2}. \quad (8)$$

122 We solve for $\lambda = \log v$, so unconstrained updates preserve $v = \exp(\lambda) > 0$. By the chain rule,
123 the corresponding efficient score is

$$S_{\lambda}^{\text{eff}} = \frac{dv}{d\lambda} S_v^{\text{eff}} = v S_v^{\text{eff}}.$$

124 Because v is positive and common within a regression, $\sum_i v S_{v,i}^{\text{eff}} = 0$ and $\sum_i S_{v,i}^{\text{eff}} = 0$ have
125 the same roots, including after plug-in substitution. The estimate is recovered as $\hat{v} = \exp(\hat{\lambda})$;
126 separate valid-root checks assess convergence and covariance validity.

127 At a candidate parameter, the residuals are $e_i = R_i - m(\mathbf{Z}_i; \beta)$. We replace expectations in
128 (6)–(8) by empirical moments and use the leave-one-out Gaussian-kernel estimators

$$\begin{aligned} \hat{f}_{-i}(e_i) &= \frac{1}{(n-1)h} \sum_{j \neq i} \phi\left(\frac{e_i - e_j}{h}\right), \\ \hat{f}'_{-i}(e_i) &= -\frac{1}{(n-1)h^3} \sum_{j \neq i} (e_i - e_j) \phi\left(\frac{e_i - e_j}{h}\right), \end{aligned} \quad (9)$$

129 where $h = 1.06 s_e n^{-1/5}$. At the same candidate value, we set

$$\hat{\mu}_3 = n^{-1} \sum_i e_i^3, \quad \hat{\mu}_4 = n^{-1} \sum_i e_i^4, \quad \hat{Q}_2 = \hat{\mu}_4 - v^2 - \hat{\mu}_3^2/v,$$

130 and substitute these quantities into (7)–(8). At the model truth, $Q_2 = \mu_4 - v^2 - \mu_3^2/v$; the
131 displayed expression is its plug-in counterpart at a candidate value, rather than the literal
132 sample mean of $q(e_i)^2$ away from the root. This deterministic normal-reference rule follows the
133 implementation in Kim (2023) and the standard density-estimation rule of Silverman (1986);
134 it also satisfies $h \rightarrow 0$ and $nh^3 \rightarrow \infty$, as required for pointwise consistency of a first-derivative
135 kernel estimator. The leave-one-out form avoids each residual contributing a kernel centered
136 on itself. For numerical stabilization, we use $\kappa_n = \min(10^{-10}, n^{-2})$ as the density floor when
137 forming $\hat{f}'_{-i}/\hat{f}_{-i}$ and reject candidates with $\hat{Q}_2 \leq \kappa_n$. At every sample size used in this article

138 $\kappa_n = 10^{-10}$, so this asymptotic formulation reproduces the reported computation exactly while
 139 allowing the safeguard to vanish as $n \rightarrow \infty$. For either regression, the estimate solves

$$\frac{1}{n} \sum_{i=1}^n \begin{pmatrix} \hat{S}_{\beta,i}^{\text{eff}} \\ \hat{S}_{\lambda,i}^{\text{eff}} \end{pmatrix} = \mathbf{0}. \quad (10)$$

140 Equations (7)–(10) specify the feasible estimator in the supplied code. Under the smoothness,
 141 tail, bandwidth and unique-root conditions of Kim (2023, Theorem 2 and its proof), together
 142 with the implementation conditions in Section S3 of Online Resource 1, the stabilized leave-one-
 143 out estimator has the same first-order expansion as their Gaussian-kernel score root. Section S3
 144 establishes this equivalence and sufficient conditions for consistent multi-start root selection,
 145 connecting the asymptotic results below to the implemented estimator.

146 2.3 The stacked mediation system

147 Let $\boldsymbol{\theta}_M = (\boldsymbol{\beta}_M^\top, \lambda_M)^\top$ and $\boldsymbol{\theta}_Y = (\boldsymbol{\beta}_Y^\top, \lambda_Y)^\top$ collect the mediator and outcome regression param-
 148 eters, including log variances. Applying the generic regression model in (5) to the response-
 149 covariate-vector pairs $(R, \mathbf{Z}) = (M, (1, T, \mathbf{X}^\top)^\top)$ and $(R, \mathbf{Z}) = (Y, (1, T, M, TM, \mathbf{X}^\top)^\top)$, re-
 150 spectively, defines the population score vectors $S_{M,i}^{\text{eff}}(\boldsymbol{\theta}_M)$ and $S_{Y,i}^{\text{eff}}(\boldsymbol{\theta}_Y)$. Each has the form
 151 $(S_{\beta}^{\text{eff}\top}, vS_v^{\text{eff}})^\top$ given in (7)–(8).

152 The scores depend on unknown error densities and design moments. For linear regressions,
 153 $\mathbb{E}\{\dot{m}_{\beta}(\mathbf{Z}; \boldsymbol{\beta})\} = \mathbb{E}(\mathbf{Z})$, estimated by sample means including $\overline{\mathbf{X}}$. These plug-in quantities
 154 support the coefficient and log-variance equations; a separate moment equation represents esti-
 155 mation of the population mean $\boldsymbol{\mu}_X$.

156 To represent estimation of both the regression parameters and the population baseline mean,
 157 define the observation-level stacked estimating function by

$$\boldsymbol{\Psi}_i(\boldsymbol{\vartheta}) = \begin{pmatrix} S_{M,i}^{\text{eff}}(\boldsymbol{\theta}_M) \\ S_{Y,i}^{\text{eff}}(\boldsymbol{\theta}_Y) \\ \mathbf{X}_i - \boldsymbol{\mu}_X \end{pmatrix}, \quad \boldsymbol{\vartheta} = (\boldsymbol{\theta}_M^\top, \boldsymbol{\theta}_Y^\top, \boldsymbol{\mu}_X^\top)^\top,$$

158 Here $\boldsymbol{\Psi}_i$ denotes the population-score version. In computation, $\hat{\boldsymbol{\Psi}}_i$ replaces each unknown
 159 density, moment, and design expectation by the sample quantities described above, and $\hat{\boldsymbol{\vartheta}}$ solves
 160 $\sum_i \hat{\boldsymbol{\Psi}}_i(\boldsymbol{\vartheta}) = \mathbf{0}$. This connects the general estimating system directly to the two regression fits

161 and to population averaging over $\mathbf{X}$.

162 The last block is the moment equation for the baseline mean:

$$\frac{1}{n} \sum_{i=1}^n (\mathbf{X}_i - \boldsymbol{\mu}_X) = \mathbf{0} \implies \hat{\boldsymbol{\mu}}_X = \overline{\mathbf{X}}.$$

163 This is the same empirical mean used in the regression scores. Its explicit inclusion in the
 164 stack carries the variability of $\overline{\mathbf{X}}$ and its covariance with the regression estimators into effect
 165 inference, without changing the regression fits.

166 For example, PNDE = $\beta_3 + \eta(\alpha_2 + \boldsymbol{\xi}_2^\top \boldsymbol{\mu}_X)$, whose first-order expansion includes $\eta \boldsymbol{\xi}_2^\top (\overline{\mathbf{X}} - \boldsymbol{\mu}_X)$.
 167 The same direct term enters TNDE and TE, but not PNIE or TNIE; it vanishes when $\eta = 0$.
 168 For effects standardized to the observed covariates, this population-mean term is omitted.

169 Define

$$\hat{A} = \frac{1}{n} \sum_{i=1}^n \frac{\partial \hat{\boldsymbol{\Psi}}_i(\hat{\boldsymbol{\vartheta}})}{\partial \boldsymbol{\vartheta}^\top}, \quad \hat{B} = \frac{1}{n} \sum_{i=1}^n \hat{\boldsymbol{\Psi}}_i(\hat{\boldsymbol{\vartheta}}) \hat{\boldsymbol{\Psi}}_i(\hat{\boldsymbol{\vartheta}})^\top.$$

170 The regression blocks of $\hat{A}$ use centered finite differences with coordinate step $d_n \max(1, |\hat{\vartheta}_j|)$,
 171 where $d_n = \min(10^{-4}, n^{-1/3})$; the baseline-mean block is $-I$. Thus $d_n = 10^{-4}$ at the re-
 172 ported sample sizes and $d_n \rightarrow 0$ asymptotically. Residuals, bandwidths, densities and moments
 173 are recomputed after each perturbation. Under Conditions (R1)–(R5) in Section S3 of On-
 174 line Resource 1, covering score regularity, consistent feasible roots, derivative convergence and
 175 covariance consistency, the estimating-equation sandwich is (van der Vaart, 1998; Tsiatis, 2006)

$$\hat{\Sigma}_{\boldsymbol{\vartheta}} = n^{-1} \hat{A}^{-1} \hat{B} \hat{A}^{-\top}.$$

176 The empirical cross-products in $\hat{B}$ retain finite-sample dependence between equations rather
 177 than forcing the off-diagonal blocks to zero.

178 Let $g(\boldsymbol{\vartheta})$ be the five-dimensional map in (3) and $G = \partial g / \partial \boldsymbol{\vartheta}^\top$. The estimated effect co-
 179 variance is $\hat{G} \hat{\Sigma}_{\boldsymbol{\vartheta}} \hat{G}^\top$. Explicit gradients are given in Section S2 of Online Resource 1 and are
 180 implemented directly in the code.

181 **Proposition 1** (Composition of the regression influence functions). *Suppose models (1)–(2)*
 182 *and Condition (R1) in Section S3 of Online Resource 1 hold, and the joint density factorizes*
 183 *into the law of $(\mathbf{X}, T)$ and the two independent regression-error densities. For $k \in \{M, Y\}$,*
 184 *let $I_k = \mathbb{E}\{S_k^{\text{eff}} S_k^{\text{eff}^\top}\}$ and $\varphi_k = I_k^{-1} S_k^{\text{eff}}$, and partition $G = (G_M, G_Y, G_X)$. Then the efficient*

185 influence function for $g(\boldsymbol{\vartheta})$ is

$$\varphi_g(O) = G_M \varphi_M(O) + G_Y \varphi_Y(O) + G_X \{\mathbf{X} - \boldsymbol{\mu}_X\}. \quad (11)$$

186 If, in addition, Conditions (R2)–(R4) hold, then the natural-effect plug-in estimator is asymp-
 187 totically linear with influence function (11) and attains the corresponding efficiency bound. Con-
 188 dition (R5) additionally ensures consistency of the propagated covariance and justifies marginal
 189 Wald inference for scalar effects with positive asymptotic variance.

190 The proposition combines regression-score efficiency with the pathwise chain rule (Kim,
 191 2023; Bickel et al., 1993; Tsiatis, 2006). Section S3 proves component orthogonality, the fitted-
 192 estimator expansion and covariance consistency. The $G_X(\mathbf{X} - \boldsymbol{\mu}_X)$ term accounts for population
 193 averaging; the vanishing numerical safeguards preserve this first-order representation under the
 194 stated conditions.

195 The reported intervals target marginal 95% coverage for each effect. The two identities in (4)
 196 imply that the joint effect covariance has rank at most three; it is propagated without inverting
 197 a five-dimensional effect covariance. A scalar Wald interval additionally requires positive first-
 198 order asymptotic variance. The regular delta approximation therefore applies away from the
 199 joint-zero point $\beta_2 = \gamma = 0$, where the PNIE gradient vanishes; inference at that nonregular
 200 point calls for a dedicated higher-order or resampling analysis.

201 2.4 Asymptotic precision and local power relative to OLS

202 The feasible-root result connects the regression efficiency theory of Kim (2023) to an explicit
 203 natural-effect variance comparison for the supplied kernel-score algorithm. For $k \in \{M, Y\}$,
 204 write the corresponding regression as $R_k = a_k + \mathbf{b}_k^\top \mathbf{W}_k + \epsilon_k$, where $\mathbf{W}_M = (T, \mathbf{X}^\top)^\top$ and $\mathbf{W}_Y =$
 205 $(T, M, TM, \mathbf{X}^\top)^\top$. Let $\boldsymbol{\omega}_k = \mathbb{E}(\mathbf{W}_k)$, $C_k = \text{Var}(\mathbf{W}_k)$, $s_k(e) = -f'_k(e)/f_k(e)$, $J_k = \mathbb{E}\{s_k(\epsilon_k)^2\}$,
 206 and

$$H_k = \begin{pmatrix} -\boldsymbol{\omega}_k^\top \\ I \end{pmatrix}.$$

207 The matrix C_k is assumed nonsingular. For natural effect g_j , let $d_{j,k} = \partial g_j / \partial \boldsymbol{\beta}_k$ be the column
 208 gradient for $\boldsymbol{\beta}_k = (a_k, \mathbf{b}_k^\top)^\top$, and let $V_{j,r}$ denote the limiting variance of $\sqrt{n}(\hat{g}_{j,r} - g_j)$ for method
 209 r .

Proposition 2 (OLS-relative asymptotic variance). *Under the regression model and Conditions (R1)–(R5) in Section S3 of Online Resource 1, for a natural effect with positive first-order variance,*

$$V_{j,\text{OLS}} - V_{j,\text{semi}} = \sum_{k \in \{M, Y\}} (v_k - J_k^{-1})(H_k^\top d_{j,k})^\top C_k^{-1} (H_k^\top d_{j,k}) \geq 0. \quad (12)$$

The inequality is strict if, for at least one regression, $v_k J_k > 1$ and $H_k^\top d_{j,k} \neq 0$.

Proof. Integration by parts gives $\mathbb{E}\{\epsilon_k s_k(\epsilon_k)\} = 1$; hence Cauchy–Schwarz gives $v_k J_k \geq 1$. Joint inversion of the coefficient and variance score blocks in (7)–(8) gives the regression-coefficient influence functions

$$\begin{aligned} \varphi_{\beta,k}^{\text{semi}} &= e_{0,k} \epsilon_k + H_k C_k^{-1} (\mathbf{W}_k - \boldsymbol{\omega}_k) \frac{s_k(\epsilon_k)}{J_k}, \\ \varphi_{\beta,k}^{\text{OLS}} &= e_{0,k} \epsilon_k + H_k C_k^{-1} (\mathbf{W}_k - \boldsymbol{\omega}_k) \epsilon_k, \end{aligned}$$

where $e_{0,k}$ selects the intercept. Independence of ϵ_k and $\mathbf{W}_k$ makes their coefficient covariance difference $(v_k - J_k^{-1}) H_k C_k^{-1} H_k^\top$. The mediator and outcome coefficient influence functions are mutually orthogonal and orthogonal to $\mathbf{X} - \boldsymbol{\mu}_X$ by conditional mean zero. The baseline-mean variance is therefore common to both methods, and applying the effect gradient yields (12). Section S3 of Online Resource 1 gives the score-block calculation. $\square$

For PNIE, the comparison simplifies to

$$V_{\text{PNIE,OLS}} - V_{\text{PNIE,semi}} = \gamma^2 (v_M - J_M^{-1}) [C_M^{-1}]_{TT} + \beta_2^2 (v_Y - J_Y^{-1}) [C_Y^{-1}]_{MM}.$$

Thus information in either error density can improve precision for an indirect effect through its corresponding regression coefficient. Equality holds for Gaussian errors, for which $J_k = 1/v_k$. This ordering characterizes asymptotic precision relative to OLS for regular effects under the independent-error model; the joint-zero PNIE point requires the separate nonregular analysis noted above. Finite-sample performance, comparisons with robust methods and properties of the confounding-sensitivity estimators are evaluated in their respective experiments.

The variance ordering also has a test-theoretic implication. For two-sided Wald tests of $g_j = 0$ at the same asymptotic level α_{test} , consider a regular local alternative $g_{j,n} = h/\sqrt{n}$ with $h \neq 0$, along which the asymptotic linear expansions and variance consistency hold. The

232 limiting power of method r is

$$\pi_{j,r}(h) = 1 - \Phi\left(z_{1-\alpha_{\text{test}}/2} - \frac{h}{\sqrt{V_{j,r}}}\right) + \Phi\left(-z_{1-\alpha_{\text{test}}/2} - \frac{h}{\sqrt{V_{j,r}}}\right),$$

233 where Φ is the standard-normal distribution function and z_p its p -quantile. As a function of
 234 $|h|/\sqrt{V_{j,r}}$, this power is increasing; hence Proposition 2 gives $\pi_{j,\text{semi}}(h) \geq \pi_{j,\text{OLS}}(h)$, strictly
 235 when (12) is strict. This comparison links variance reduction to local power for regular alterna-
 236 tives at a common asymptotic test size and motivates the independently calibrated finite-sample
 237 evaluation below.

238 2.5 Deterministic multi-start computation

239 Let $(\tilde{\beta}, \tilde{v})$ be the OLS pilot from the same regression and let

$$d_j = \max\{0.05, c \max(1, |\tilde{\beta}_j|)\}, \quad c = 0.10.$$

240 The lower bound prevents an exactly zero or small OLS coefficient from producing indistin-
 241 guishable starts; scaling by $\max(1, |\tilde{\beta}_j|)$ makes the perturbation relative for larger coefficients.
 242 With $\mathbf{s} = (1, -1, 1, -1, \dots)^\top$, the seven starts are

$$\begin{aligned} &(\tilde{\beta}, \log \tilde{v}), (\tilde{\beta} + \mathbf{d}, \log \tilde{v}), (\tilde{\beta} - \mathbf{d}, \log \tilde{v}), \\ &(\tilde{\beta} + \mathbf{s} \circ \mathbf{d}, \log \tilde{v}), (\tilde{\beta} - \mathbf{s} \circ \mathbf{d}, \log \tilde{v}), (\tilde{\beta}, \log(1.25\tilde{v})), (\tilde{\beta}, \log(0.80\tilde{v})). \end{aligned}$$

243 The all-positive, all-negative, and alternating perturbations explore distinct coefficient direc-
 244 tions without random restarts; the final two starts separately probe the variance coordinate
 245 and are symmetric on the log-variance scale because $0.80 = 1/1.25$. With the reported values
 246 of c , the 0.05 floor is redundant because $c \max(1, |\tilde{\beta}_j|) \geq 0.05$ and becomes active only in imple-
 247 mentations using a smaller c . These rules are designed to make root exploration deterministic,
 248 transparent and reproducible.

249 Each start uses `rootSolve::multiroot()` with at most 200 iterations and relative and
 250 absolute tolerance $a_n = \min(10^{-9}, n^{-1})$. Valid roots require finite parameters, maximum abso-
 251 lute mean score below $b_n = \min(10^{-6}, n^{-1})$, Jacobian reciprocal condition number above 10^{-10}
 252 and finite covariance with positive diagonal. At the reported sample sizes, $a_n = 10^{-9}$ and
 253 $b_n = 10^{-6}$; asymptotically, $b_n = o(n^{-1/2})$ supplies the accuracy in Condition (R2). Duplicate

254 roots are removed after rounding parameters to seven decimal places. Section S4 gives the
 255 complete algorithm.

256 If more than one valid root remains, we choose the one minimizing

$$D(\boldsymbol{\theta}) = \frac{(\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}})^\top (\mathbf{Z}^\top \mathbf{Z}/n)(\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}})}{\tilde{v}} + \{\lambda - \log(\tilde{v})\}^2.$$

257 The first term measures the change in fitted linear predictor on the observed design scale,
 258 while the second compares log variances. Under the correctly specified linear mean and zero
 259 conditional error mean, OLS provides a consistent coefficient pilot for non-Gaussian errors and
 260 a principled anchor for root selection. OLS supplies initialization and the selection reference;
 261 all reported proposed-method estimates solve the density-score equations. We repeat all starts
 262 with $c \in \{0.05, 0.10, 0.20\}$ in the algorithm-sensitivity study.

263 **Use of generative AI.** OpenAI Codex was used to assist with manuscript drafting and revi-
 264 sion, mathematical derivations and exposition, and the development and debugging of analysis
 265 code. The author takes full responsibility for the mathematical arguments, code, numerical
 266 results, interpretations and final manuscript.

267 3 Simulation studies

268 3.1 Data-generating mechanisms

269 For the main experiment, $X \sim N(0, 1)$ and $T \sim \text{Bernoulli}(0.5)$ independently, with

$$M = 0.2 + 0.4T + 0.3X + \epsilon_M,$$

$$Y = 0.5T - 0.8M + 1.0TM + 0.4X + \epsilon_Y.$$

270 The errors are independently drawn from the same standardized distribution. The true effect
 271 vector (PNIE, TNIE, PNDE, TNDE, TE) is $(-0.32, 0.08, 0.70, 1.10, 0.78)$.

272 Four zero-mean, unit-variance errors are considered. Let ϕ and Φ denote the standard
 273 normal density and distribution function. For each nonstandard raw variable U below, we
 274 generate $\epsilon = (U - \mu_U)/\sigma_U$; equivalently, its final density is $f_\epsilon(e) = \sigma_U f_U(\mu_U + \sigma_U e)$.

275 1. Gaussian: $f_\epsilon(e) = \phi(e)$.

2. Skew-normal: $f_U(u) = 2\phi(u)\Phi(6u)$. With $\delta = 6/\sqrt{37}$, the exact constants are $\mu_U = \delta\sqrt{2/\pi} = 0.7870$ and $\sigma_U = \sqrt{1 - 2\delta^2/\pi} = 0.6169$.

3. Asymmetric mixture:

$$f_U(u) = \frac{0.82}{0.35}\phi\left(\frac{u+0.7}{0.35}\right) + \frac{0.18}{0.75}\phi\left(\frac{u-3.2}{0.75}\right),$$

with $\mu_U = 0.002$ and $\sigma_U = 1.5642$.

4. Symmetric bimodal mixture:

$$f_U(u) = \frac{0.5}{0.45}\phi\left(\frac{u+1.7}{0.45}\right) + \frac{0.5}{0.45}\phi\left(\frac{u-1.7}{0.45}\right),$$

with $\mu_U = 0$ and $\sigma_U = \sqrt{0.45^2 + 1.7^2} = 1.7586$.

The densities are shown in Figure S1 of Online Resource 1.

The main experiment uses $n \in \{200, 500\}$ and 1,000 Monte Carlo replications in every cell. The smaller size probes finite-sample behavior of the estimated density derivative, while the larger size assesses stabilization at a still-moderate sample size. We report bias, root mean squared error (RMSE), 95% Wald coverage, average interval length, numerical success rate, and the Monte Carlo standard error (MCSE) of coverage. Failures remain in the denominator of the reported success rate but are excluded from performance summaries, whose effective replication counts are reported in the complete tables.

A separate comparator experiment uses $n = 300$, intermediate between the two main-study sizes, and 1,000 Monte Carlo replications per error distribution. The four distributions yield 4,000 data sets and 20,000 attempted method fits. The experiment compares OLS delta intervals and the proposed method with (i) the quasi-Bayesian approximation of Imai et al. (2010a,b) using 1,000 draws, (ii) their nonparametric percentile bootstrap using 499 resamples, both through the R package `mediation` (Tingley et al., 2014), and (iii) an imputation-based natural effect model with sandwich standard errors through the R package `medflex` (Steen et al., 2017). All procedures target the same five effects and use compatible linear mean structures. For `medflex`, outcome imputation uses $Y \sim T * M + T * X$, and the natural effect model uses $Y \sim T_0 * T_1 + T_0 * X$ with centered X . The $T_0 X$ term is required by the interaction-induced covariate dependence of the natural direct effects. The additional TX imputation term satisfies the package's working-model compatibility check; its true coefficient is zero in the simulation. Thus

Table 1 Main Monte Carlo results based on 1,000 attempts per cell and summarized over valid fits; counts and success rates appear in Table S1. G, Gaussian; SN, skew-normal; AM, asymmetric mixture; BM, symmetric bimodal. Boldface marks lower RMSE and shorter length within each design and effect; these descriptive highlights must be interpreted jointly with coverage and numerical success. Bias is signed

<i>n</i>	Error	Method	PNIE				TE			
			Bias	RMSE	Cov.	Length	Bias	RMSE	Cov.	Length
200	G	OLS delta	0.003	0.121	0.947	0.467	-0.001	0.162	0.949	0.643
200	G	Proposed	0.004	0.130	0.932	0.495	-0.001	0.175	0.938	0.682
200	SN	OLS delta	0.005	0.122	0.945	0.469	0.000	0.171	0.937	0.645
200	SN	Proposed	0.005	0.100	0.951	0.388	0.000	0.144	0.936	0.553
200	AM	OLS delta	0.001	0.126	0.942	0.475	0.008	0.168	0.946	0.646
200	AM	Proposed	-0.001	0.055	0.940	0.120	0.002	0.088	0.937	0.327
200	BM	OLS delta	0.004	0.119	0.940	0.470	0.002	0.157	0.957	0.646
200	BM	Proposed	0.000	0.034	0.929	0.121	0.002	0.110	0.950	0.327
500	G	OLS delta	0.001	0.074	0.956	0.297	-0.006	0.107	0.946	0.409
500	G	Proposed	0.001	0.078	0.958	0.311	-0.005	0.111	0.950	0.424
500	SN	OLS delta	0.000	0.077	0.955	0.298	-0.001	0.109	0.948	0.409
500	SN	Proposed	0.004	0.059	0.942	0.233	-0.001	0.087	0.944	0.337
500	AM	OLS delta	-0.004	0.075	0.946	0.299	0.007	0.106	0.947	0.409
500	AM	Proposed	0.000	0.019	0.949	0.074	0.002	0.054	0.948	0.205
500	BM	OLS delta	-0.003	0.076	0.945	0.298	-0.003	0.101	0.955	0.409
500	BM	Proposed	0.000	0.019	0.959	0.076	-0.001	0.053	0.941	0.206

this comparator has a slightly richer outcome working model, as detailed in Online Resource 1.

The two `mediation` entries are inferential options within its *model-based* approach. The package’s single-experiment design-based routine is formulated for a discrete mediator, and its parallel-design routine is formulated for mediator manipulation (Tingley et al., 2014). The model-based routines therefore provide the like-for-like comparison for the continuous mediator and target used here.

The OLS delta analysis uses an empirical estimating-equation sandwich for the two regressions and the baseline mean, providing likelihood-free covariance estimation based on first two moments. The proposed estimator additionally uses the shape of a non-Gaussian error density. All procedures target the same population effect definitions, with method-specific uncertainty calculations: quasi-Bayesian simulation conditions on observed covariates, the `medflex` decomposition uses package-native standard errors evaluated at the centered covariate reference, the stacked calculation adds baseline-mean variability, and the nonparametric bootstrap captures that variability through observation-level resampling.

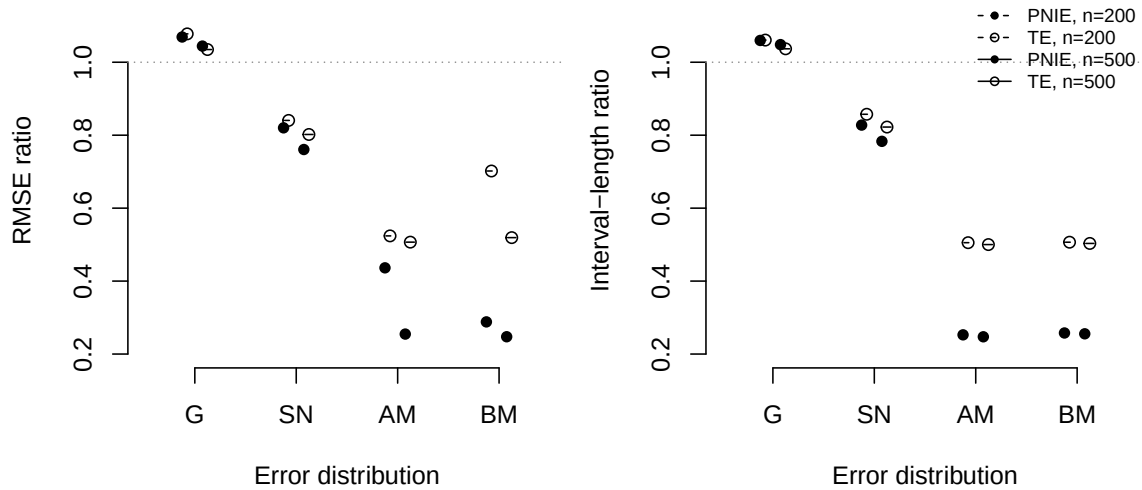


Fig. 1 Ratios of proposed-method RMSE and average 95% interval length to their OLS counterparts, using method-specific summaries over numerically valid fits. G, SN, AM, and BM denote Gaussian, skew-normal, asymmetric-mixture, and symmetric-bimodal errors; values below one favor the proposed method. Point shape identifies PNIE or TE, and line type identifies sample size

3.2 Main and comparator results

Table 1 reports PNIE and TE; Table S1 of Online Resource 1 gives all five effects and diagnostics. Across the 30 non-Gaussian sample-size-by-effect cells, proposed-to-OLS RMSE ratios ranged from 0.245 to 0.841 and interval-length ratios from 0.247 to 0.857. Proposed coverage ranged from 0.918 to 0.959. Numerical success was 1.000 under the skew-normal error, 0.887 and 0.921 for the asymmetric and bimodal mixtures at $n = 200$, and 0.999 and 1.000 for those mixtures at $n = 500$.

The Gaussian setting provides the reference in which OLS attains its efficiency benchmark. There, the proposed-to-OLS RMSE ratios were 1.035–1.078 and interval-length ratios were 1.037–1.063, showing slightly larger RMSE and interval lengths for the feasible density-score estimator at this benchmark.

On common successful data sets, the 30 non-Gaussian main-study cells gave paired RMSE ratios of 0.245–0.841 and interval-length ratios of 0.247–0.857, confirming the same precision pattern (Table S3). The fraction of *all attempted data sets* returning a proposed interval that covered the truth ranged from 0.828–0.959. This report-and-cover measure combines numerical success with conditional coverage; Section S5 gives its definition and the paired-ratio Monte Carlo standard errors.

Table 2 Comparator study at $n = 300$ based on 1,000 attempts per cell, summarized over valid fits. The Imai procedures use 1,000 quasi-Bayesian draws or 499 bootstrap resamples per data set. Error abbreviations are as in Table 1. Boldface marks the smallest displayed RMSE and shortest length within each design and effect; these descriptive highlights are interpreted jointly with coverage and numerical success

Error	Method	PNIE			TE		
		RMSE	Cov.	Length	RMSE	Cov.	Length
G	OLS delta	0.100	0.952	0.383	0.135	0.955	0.527
G	Imai quasi-Bayes	0.100	0.950	0.387	0.135	0.962	0.573
G	Imai bootstrap	0.100	0.946	0.384	0.147	0.966	0.565
G	medflex NEM	0.100	0.950	0.385	0.135	0.954	0.522
G	Proposed	0.107	0.942	0.406	0.139	0.948	0.552
SN	OLS delta	0.099	0.950	0.385	0.133	0.952	0.527
SN	Imai quasi-Bayes	0.100	0.949	0.388	0.133	0.970	0.572
SN	Imai bootstrap	0.099	0.939	0.386	0.146	0.968	0.566
SN	medflex NEM	0.099	0.950	0.386	0.133	0.950	0.522
SN	Proposed	0.079	0.955	0.310	0.112	0.947	0.443
AM	OLS delta	0.101	0.944	0.387	0.141	0.935	0.527
AM	Imai quasi-Bayes	0.102	0.944	0.389	0.141	0.954	0.573
AM	Imai bootstrap	0.101	0.945	0.390	0.153	0.953	0.567
AM	medflex NEM	0.102	0.948	0.388	0.140	0.935	0.522
AM	Proposed	0.026	0.938	0.096	0.069	0.944	0.264
BM	OLS delta	0.096	0.956	0.385	0.133	0.951	0.527
BM	Imai quasi-Bayes	0.096	0.956	0.388	0.134	0.960	0.574
BM	Imai bootstrap	0.096	0.957	0.386	0.148	0.961	0.566
BM	medflex NEM	0.096	0.955	0.386	0.134	0.948	0.522
BM	Proposed	0.029	0.949	0.098	0.079	0.941	0.266

Each point in Figure 1 is the proposed-to-OLS ratio for one error distribution, sample size and effect; a value of 0.5 means half the RMSE or average interval length of OLS. The figure uses method-specific valid fits, complemented by the common-data comparison above. Interval length is interpreted jointly with coverage and numerical success.

Table 2 extends the evaluation to the package-based comparators. Among the 15 non-Gaussian distribution-by-effect cells, the proposed method had the smallest RMSE in all 15, with coverage from 0.924 to 0.959. Its numerical success was 1.000, 0.962, and 0.978 under the skew-normal, asymmetric-mixture, and bimodal errors, respectively; all four comparators succeeded in every attempted fit. Under Gaussian errors, RMSE was comparable to or slightly larger than that of the comparators. The two **mediation** procedures quantify uncertainty differently, while **medflex** estimates a natural effect model; the proposal additionally uses an estimated regression-error density within the model of Section 2.1.

3.3 Null size, power, and numerical sensitivity

For null-size and power assessment, we use $n = 300$, standardized asymmetric-mixture errors, $\gamma = -0.26$, $\eta = 0.8$ and $\beta_2 \in \{0, 0.1, 0.2, 0.3, 0.4\}$. The true PNIE is $-0.26\beta_2$, with the null at $\beta_2 = 0$. Each setting has 1,000 attempted replications of the two-sided test of $H_0 : \text{PNIE} = 0$. At the nominal 5% level, rejection occurs when the 95% Wald interval excludes zero, equivalently when $|\widehat{\text{PNIE}}/\widehat{\text{SE}}| > z_{0.975} \approx 1.96$. This standard-normal reference concerns the test statistic, not a Gaussian assumption for the regression errors.

Let $\mathcal{V}$ be the set of numerically valid fits and $[L_r, U_r]$ the interval in replication r . The empirical rejection probability and its binomial Monte Carlo standard error are

$$\hat{p}_{\text{rej}} = \frac{1}{|\mathcal{V}|} \sum_{r \in \mathcal{V}} \mathbf{1}\{L_r > 0 \text{ or } U_r < 0\}, \quad \text{MCSE}(\hat{p}_{\text{rej}}) = \sqrt{\frac{\hat{p}_{\text{rej}}(1 - \hat{p}_{\text{rej}})}{|\mathcal{V}|}}.$$

At $\beta_2 = 0$, rejection estimates the conditional type I error rate: closeness to the nominal 0.05, rather than a larger value, is desirable. At $\beta_2 \neq 0$, it estimates power among valid fits, so higher rejection is preferable at comparable type I error rates. Coverage measures containment of the true effect; rejection equals one minus coverage only at the null, using the same valid fits.

The original nominal experiment gave null rejection proportions among valid fits of 0.016 for OLS and 0.056 for the proposal (Table S4), indicating conservative OLS rejection and closer-to-nominal proposed rejection. For power comparisons at similar null sizes, we calibrate method-specific cutoffs to 5% using 5,000 independent null data sets and freeze them for evaluation on 1,000 new data sets per effect value. The OLS cutoff decreases from 1.96 to 1.669, easing rejection, while the proposed cutoff is nearly unchanged at 1.967 (Table S5). The held-out null checks attained size; Section S6 details the order-statistic rule and two-stage Monte Carlo uncertainty.

The comparison includes Huber-FIX and Huber-SEL, motivated by Wang et al. (2025), using the same regressions and PNIE slope-product target. Huber-FIX uses 1.345 times a LAD-based residual scale; Huber-SEL minimizes an estimated variance factor over the authors' tuning grid. Under independent errors, the intercept absorbs the Huber location shift, preserving the slope target even for asymmetric errors. Section S6 derives this result and the interaction-adapted covariance; Tables S5–S7 give cutoffs, held-out rejection and matched-data contrasts.

Figure 2 compares nominal and calibrated rejection on the same new evaluation samples. These are distinct from the original samples in Table S4, so the nominal rejection proportions

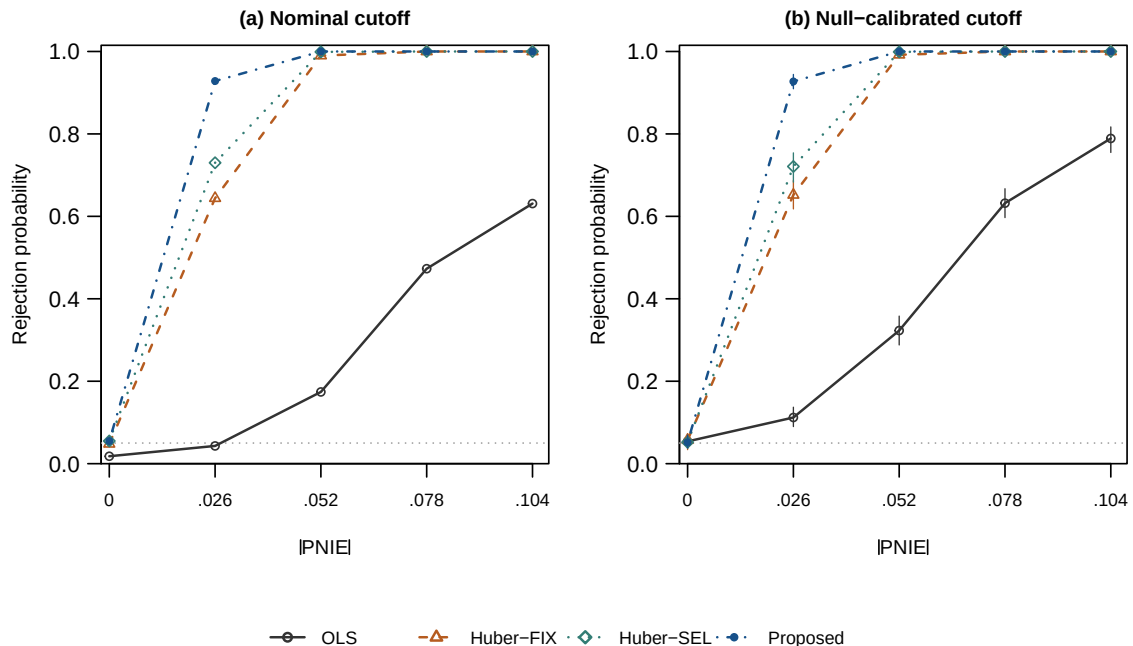


Fig. 2 Held-out PNIE rejection probabilities under the asymmetric-mixture error at $n = 300$. Panel (a) uses the common normal cutoff; panel (b) uses method-specific cutoffs from 5,000 independent null-calibration data sets. Each point uses 1,000 new evaluation data sets, shared by the four methods. The ordinate is type I error at $|\text{PNIE}| = 0$ and power elsewhere, conditional on a valid fit. Vertical segments in (b) are pointwise 95% Monte Carlo uncertainty intervals from resampling both simulation stages; the horizontal line marks 0.05

need not coincide exactly. After calibration, held-out null rejection is 0.054 for OLS, 0.057 for Huber-FIX, 0.052 for Huber-SEL and 0.052 for the proposal. At $\beta_2 = 0.1$, corresponding to $\text{PNIE} = -0.026$, calibrated conditional power is 0.112, 0.652, 0.721 and 0.927, respectively. Thus the weak-effect power gain persists after addressing OLS conservatism and extends to these Huber comparators in the same design. On data sets where all four procedures succeed, the proposed-minus-Huber-SEL power difference is 0.204 with a two-stage Monte Carlo 95% interval of $[0.172, 0.238]$.

Proposed numerical success ranges from 0.960–0.967. At $\text{PNIE} = -0.026$, the all-attempt fraction returning a valid fit and rejecting is 0.892, exceeding both Huber rates. For larger effects, near-unit Huber conditional power and complete numerical success yield higher all-attempt detection, locating the additional advantage at the weak effect. The calibration diagnoses this DGP-specific comparison; applications use the working-model inference above. Because $\gamma = -0.26$ at the null, the experiment concerns the regular one-component null, with the joint-zero case requiring separate nonregular analysis.

Using 300 common data sets for each mixture, we also fit starting-value scales $c = 0.05, 0.10$,

and 0.20. Success was 0.957, 0.900, and 0.860 for the asymmetric mixture and 0.950, 0.940, and 0.910 for the bimodal mixture as c increased. Mean PNIE stayed between -0.321 and -0.318 , and valid-root counts averaged near one. Table S8 shows stable effect estimates among valid fits alongside lower success at wider perturbations; unsuccessful fits were recorded as failures rather than replaced by OLS estimates.

3.4 Sensitivity to variance misspecification

We separately examine departures from the independent-error restriction without changing the conditional means or the causal identification conditions. At $n = 300$, let $\epsilon_M = s(T, X)U_M$ and $\epsilon_Y = s(T, X)U_Y$, where the standardized innovations are mutually independent and independent of (T, X) . We use Gaussian and asymmetric-mixture innovations from the main study. The two variance mechanisms are

$$s_X^2(X) = \exp(\kappa X - \kappa^2/2), \quad s_T^2(T) = \frac{\exp\{\kappa(2T - 1)\}}{\cosh(\kappa)},$$

with $\kappa \in \{0, 0.15, 0.30, 0.60\}$. Both satisfy $\mathbb{E}(s^2) = 1$, preserving marginal error variance. Sharing the $\kappa = 0$ reference gives 14 configurations, each with 500 attempts: 7,000 data sets and 14,000 method fits. OLS uses its stacked sandwich; the proposed estimator and numerical settings are unchanged. Section S7 provides the structural identification argument, variance contrasts and Monte Carlo uncertainty.

The results distinguish the two variance mechanisms. Under asymmetric-mixture innovations with covariate-dependent variance, proposed PNIE coverage was 0.959, 0.963 and 0.948 as κ increased, with numerical success of 0.920 at $\kappa = 0.60$. Treatment-dependent variance produced PNIE biases of 0.026, 0.065 and 0.180 and conditional coverages of 0.802, 0.327 and 0.000; corresponding OLS coverages were 0.952, 0.954 and 0.952. At the strongest treatment-dependent setting, proposed numerical success was 0.798 and total-effect coverage was 0.393. Coverage departures are evident at a treatment-to-control variance ratio of about 1.35 ($\kappa = 0.15$) in this skewed setting. The paired-fit comparisons show the same pattern on common retained data sets. These results provide a practical reason to assess treatment-group variance differences alongside error shape and identify covariate-dependent error distributions as a priority for extending the homoscedastic framework. Table S9 and Figure S2 document these calibration changes under preserved causal identification but misspecified error variances.

4 Density-score confounding-sensitivity analysis

We extend the density-score strategy of Section 2 to residual-correlation sensitivity analysis. Allowing structural outcome errors to depend on M prevents direct reuse of the outcome-regression score under its original independence assumption. We therefore use reduced-form regressions with unspecified marginal error densities and propagate their joint uncertainty through the sensitivity functional.

4.1 Sensitivity target and reduced-form model

We retain treatment exchangeability, consistency, positivity and a linear structural outcome model, while allowing mediator–outcome confounding. The pooled mediator and treatment-specific reduced-outcome means are

$$\mathbb{E}(M \mid T, \mathbf{X}) = \alpha_2 + \beta_2 T + \boldsymbol{\xi}_2^\top \mathbf{X}, \quad q_t(\mathbf{X}) = \mathbb{E}(Y \mid T = t, \mathbf{X}) = a_t + \mathbf{d}_t^\top \mathbf{X}.$$

Let $U = M - \mathbb{E}(M \mid T, \mathbf{X})$ be independent of $(T, \mathbf{X})$ and let $W_t = Y - q_t(\mathbf{X})$ within $T = t$. The joint law of (U, W_t) is assumed not to depend on $\mathbf{X}$ within that stratum, while dependence between U and W_t is allowed. These restrictions permit marginal density-score regression without assuming independence between regression blocks. Write $v_M = \text{Var}(U)$, $v_t = \text{Var}(W_t \mid T = t)$ and $\omega_t = \text{Cov}(U, W_t \mid T = t)$, with $v_M v_t > \omega_t^2$. The mediator’s error law is common across treatment groups; outcome reduced-form error laws may differ. Together with the linear means, these assumptions define the reduced-form working model.

For a structural mediator slope b_t in treatment stratum t , the structural outcome error is $E_t = W_t - b_t U$. Specify $\rho_t = \text{Corr}(U, E_t \mid T = t)$ as a sensitivity parameter. Solving the covariance identities gives

$$b_t(\rho_t) = \frac{\omega_t}{v_M} - \frac{\rho_t}{\sqrt{1 - \rho_t^2}} \sqrt{\frac{v_t}{v_M} - \frac{\omega_t^2}{v_M^2}}, \quad -1 < \rho_t < 1. \quad (13)$$

Equation (13) is the stratum-specific identity of Imai et al. (2010a, Section 5 and Appendix D). Because it uses only second-moment identities, it requires no Gaussian error assumption and applies to the stated semiparametric model with unspecified error densities. Its causal interpretation requires an additive outcome error invariant to mediator intervention and a constant mediator slope within each stratum; $b_1 - b_0$ represents treatment–mediator interaction. Sec-

tion S8 of Online Resource 1 gives the structural derivation.

Let $\tau = (a_1 - a_0) + (\mathbf{d}_1 - \mathbf{d}_0)^\top \boldsymbol{\mu}_X$. The sensitivity functionals are

$$\begin{aligned}\delta_t(\rho_t) &= \beta_2 b_t(\rho_t), \\ \zeta_0(\rho_1) &= \tau - \delta_1(\rho_1), \quad \zeta_1(\rho_0) = \tau - \delta_0(\rho_0).\end{aligned}\tag{14}$$

The illustrations follow the analyst-specified path $\rho_0 = \rho_1 = \rho$ through the two-correlation space; these correlations are not identified from observed data alone. At the population level, $\rho = 0$ recovers the usual linear mean-effect decomposition, but zero correlation need not imply full sequential ignorability with non-Gaussian errors.

4.2 Semiparametric estimation and joint inference

To estimate (14), we apply the density scores of Section 2.2 to M on $(1, T, \mathbf{X})$ and to Y on $(1, \mathbf{X})$ within each treatment group, with an unspecified marginal error density for each regression. The OLS reduced-form benchmark uses the same regressions and an empirical sandwich, with residual second-moment equations for the variances. Both implementations append $1(T = t)(UW_t - \omega_t)$ for $t = 0, 1$ and $\mathbf{X} - \boldsymbol{\mu}_X$ to their stacked equations. Cross-product moments and cross-block empirical score covariance account for dependence between the reduced-form errors. Full equations, bread derivatives and effect gradients are given in Section S8 of Online Resource 1.

Write the resulting nuisance vector as θ and the map in (14) as $g(\theta; \rho)$. The sensitivity estimator is $g(\hat{\theta}; \rho)$. For fixed ρ , if the fitted marginal regressions and appended moments satisfy the joint asymptotic-linearity conditions stated in Section S8, then the ordinary delta method gives

$$\varphi_{g, \rho} = G_\rho \varphi_\theta, \quad \widehat{\text{Var}}\{g(\hat{\theta}; \rho)\} = \hat{G}_\rho \widehat{\text{Var}}(\hat{\theta}) \hat{G}_\rho^\top.$$

We call the implementations Ordinary Least Squares Reduced-Form (OLS-RF) and Density-Score Reduced-Form (DS-RF). Efficient marginal scores under the working model supply the DS-RF components, with mediator pooling exploiting the common error shape to estimate β_2 .

For $\beta_2 \neq 0$, solving the sensitivity identity for the point-estimate zero boundary gives

$$\rho_t^\dagger = \omega_t / \sqrt{v_M v_t}.$$

Its uncertainty uses the full nuisance covariance and a Fisher-transform delta interval. We distinguish the point-estimate zero $\hat{\rho}_t^\dagger$ from the correlation where a pointwise effect interval first includes zero, starting with the interval at $\rho = 0$. The former has a common first-order residual-moment representation for both methods; density-score gains in curve estimation arise primarily through regression coefficients and their joint covariance.

4.3 Numerical evaluation

The extension is evaluated at $n = 300$, with 500 attempts per configuration, using Gaussian and asymmetric-mixture innovations and $\rho_{\text{true}} \in \{-0.3, 0, 0.3\}$. The original structural means are retained. Independent standardized innovations generate $\epsilon_M = U_M$ and $\epsilon_Y = \rho_{\text{true}}U_M + \sqrt{1 - \rho_{\text{true}}^2}U_Y$, preserving natural effects while introducing confounding at nonzero correlation. Both methods are evaluated at $\rho = \rho_{\text{true}}$ and the unadjusted reference $\rho = 0$. The six configurations yield 3,000 data sets and 6,000 method attempts. Table S10 gives representative effects at the true correlation; Section S8 details verification and complete-output reporting.

At the correct assumed correlation, DS-RF succeeds in 498, 500 and 499 of 500 asymmetric-mixture attempts at $\rho_{\text{true}} = -0.3, 0, 0.3$, respectively; all Gaussian and OLS-RF attempts succeed. On paired successful mixture samples, the PNIE RMSE ratios (DS-RF/OLS-RF) are 1.011, 0.706 and 0.559, and the interval-length ratios are 1.014, 0.698 and 0.524. DS-RF PNIE coverage is 0.958, 0.952 and 0.946. These results identify how the PNIE precision gain varies with the generating correlation. TNIE RMSE ratios are 0.794, 0.712 and 0.746, with DS-RF coverage between 0.928 and 0.944; TNDE RMSE is higher at the two lower correlations. Total-effect precision and zero-boundary interval lengths are nearly unchanged, and Gaussian results are consistent with the OLS-efficient reference. These patterns connect effect-specific precision gains to curve estimation and distinguish them from estimation of the moment-based zero boundary.

Evaluation at the unadjusted reference $\rho = 0$ quantifies the consequences of leaving confounding unaccounted for: for mixture errors at true correlations -0.3 and 0.3 , DS-RF PNIE biases are -0.119 and 0.120 , and coverage is 0.512 and 0.158 . This comparison highlights the complementary roles of nuisance-estimation precision and specification of the sensitivity value in recovering the causal effect. Monte Carlo uncertainty for the 500 attempts is quantified through complete effect-specific standard errors; paired summaries, failures and all-attempted

report-and-cover fractions are also supplied in the project repository.

4.4 Performance along sensitivity curves

We evaluate the sensitivity curves at $\rho = -0.8, -0.75, \dots, 0.8$, using the same 500 data sets per configuration and identical estimator settings. Curve evaluations remain correlated within each replication; the Monte Carlo sample size is 500, not the number of grid evaluations.

The evaluation uses two complementary targets. If the generating correlation is r , the population PNIE/TNIE sensitivity curves in this design are

$$\delta_t^{(r)}(\rho) = 0.4 \left\{ -0.8 + t + r - \frac{\rho}{\sqrt{1 - \rho^2}} \sqrt{1 - r^2} \right\}. \quad (15)$$

Curve bias, RMSE and pointwise coverage target (15); inclusion of the fixed generating effect is recorded separately. The targets coincide at $\rho = r$, distinguishing curve calibration from causal recovery when a different correlation is assumed.

Figure 3 displays PNIE curve estimates and sampling ranges. Figures S3–S4 and Table S11 add pointwise coverage, paired RMSE and zero-boundary inference; the project repository retains all effects, correlations, Monte Carlo errors and all-attempted summaries.

Across all three generating correlations and the 33-point grid, paired PNIE RMSE ratios range from 0.985 to 1.054 under Gaussian innovations and from 0.341 to 1.363 under asymmetric-mixture innovations. The asymmetric-mixture results identify regions with substantial precision gains alongside regions with comparable or higher RMSE. Mixture TNIE ratios similarly range from 0.367 to 1.085. This pattern is explained by the nuisance components that dominate the effect gradient: near a zero of $b_t(\rho)$, the derivative of δ_t with respect to β_2 is small, whereas regions with a larger derivative can benefit more from precise mediator treatment-effect estimation.

For mixture innovations, DS-RF pointwise curve coverage ranges from 0.924 to 0.966 for PNIE and from 0.904 to 0.964 for TNIE, quantifying the calibration accompanying the precision pattern across the grid. Gaussian coverage likewise varies with the assumed correlation. Zero-boundary RMSE and interval lengths are nearly identical between methods (Table S11), as predicted by their shared first-order residual-moment representation. Together, the results characterize three complementary features of sensitivity analysis: precision along the assumed-correlation curve, pointwise interval calibration and the point-estimate zero boundary. The separately retained generating-effect inclusion frequencies quantify how recovery of the causal

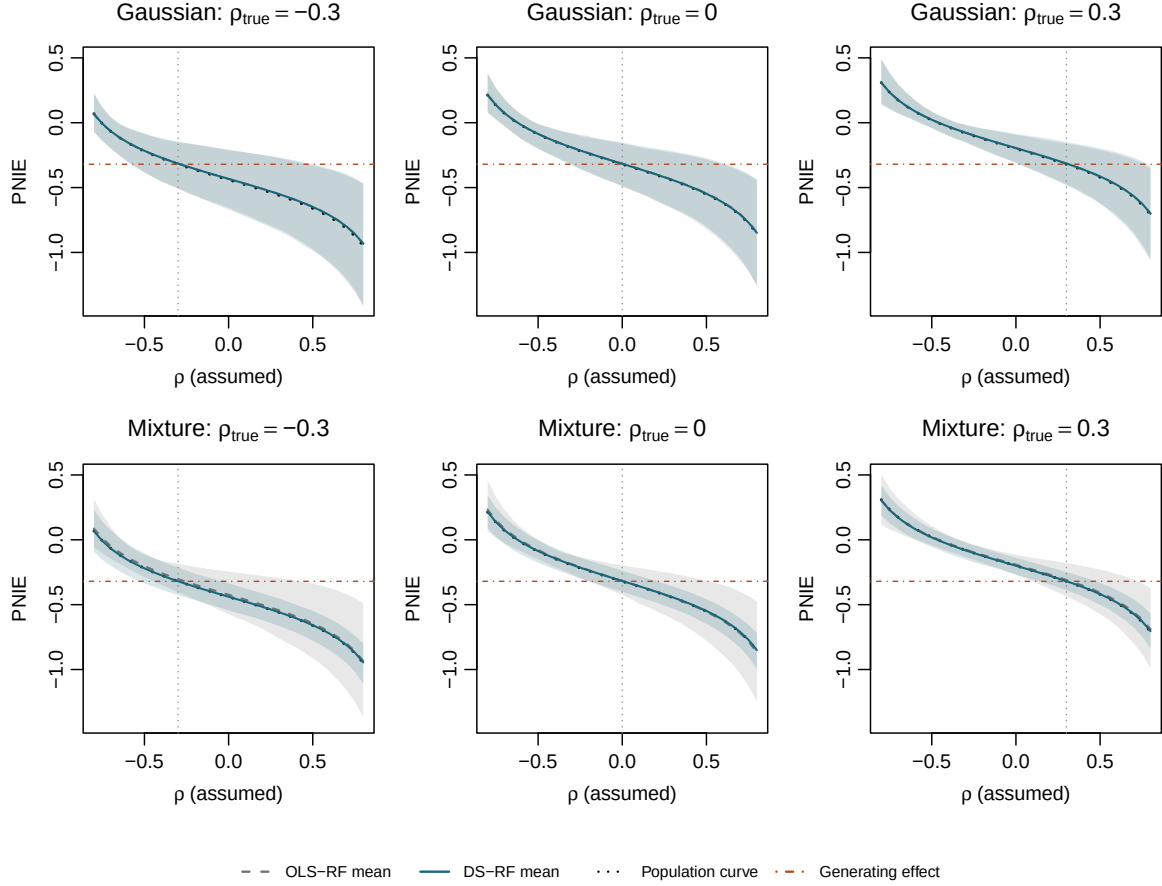


Fig. 3 Simulated PNIE sensitivity curves at $n = 300$, with 500 attempted replications for each generating correlation and innovation density. Lines show Monte Carlo mean estimates and the population sensitivity curve. Gray and blue shading show central 90% empirical sampling ranges of the OLS-RF and DS-RF estimates across successful replications, respectively. The horizontal reference is the generating PNIE, -0.32 , and the vertical reference is the generating correlation. All six configurations are shown.

effect depends on the specified confounding value.

5 JOBS II behavioral-intervention illustration

The JOBS II data distributed with version 4.5.1 of the R package `mediation` contain 899 complete observations from a randomized intervention designed to strengthen job-search skills and protect mental health among unemployed workers (Vinokur and Schul, 1997; Imai et al., 2010b; Tingley et al., 2014). Treatment is randomized assignment to the intervention, the mediator is job-search self-efficacy, and the outcome is post-intervention depressive symptoms. We use the package sample as a methodological illustration of natural-effect estimation and sensitivity analysis.

Both mean models adjust for pretreatment depressive symptoms, economic hardship, sex and age, and the outcome model includes treatment–mediator interaction. Every package observation is complete for these variables and is retained. Effects are standardized to the empirical baseline-covariate distribution. All five procedures use the same 899 observations. The two analyses implemented in the R package `mediation` use 5,000 quasi-Bayesian draws and 1,999 percentile-bootstrap resamples, respectively; the R package `medflex` supplies the natural effect model.

Table 3 reports the proposed estimates, while Figure 4 compares all five methods; their complete numerical results appear in Table S12 of Online Resource 1. The four conventional analyses give similar small negative estimates. OLS gives PNIE -0.0136 with nominal 95% interval $[-0.0360, 0.0087]$, whereas the proposal gives -0.0109 with $[-0.0205, -0.0013]$. The corresponding TNIE estimates are -0.0097 and -0.0072 . The proposed-to-OLS standard-error ratios are 0.430 for PNIE, 0.437 for TNIE, and 0.754–0.774 for the direct and total effects. Thus the largest precision difference occurs for the indirect effects, whose proposed nominal intervals exclude zero under the working model.

Section S9 supplies the complete estimates and diagnostics. OLS and the proposal target the same effect definitions through different estimating equations; under misspecification, their limiting targets may also differ. The empirical comparison is therefore interpreted with diagnostics, while comparative bias and efficiency are assessed against known targets in the simulations.

The default fit selected the OLS-centered start in both equations, with one distinct valid root and mean-score norms below 3×10^{-12} . Scales $c = 0.05, 0.10, 0.20$ gave identical displayed results

Table 3 Proposed density-score estimates of JOBS II natural effects and nominal 95% limits ($n = 899$). Negative values indicate lower post-intervention depressive symptoms under assignment to the intervention

Effect	Estimate	Lower	Upper
PNIE	-0.0109	-0.0205	-0.0013
TNIE	-0.0072	-0.0144	-0.0001
PNDE	-0.0272	-0.0892	0.0349
TNDE	-0.0235	-0.0835	0.0364
TE	-0.0344	-0.0962	0.0274

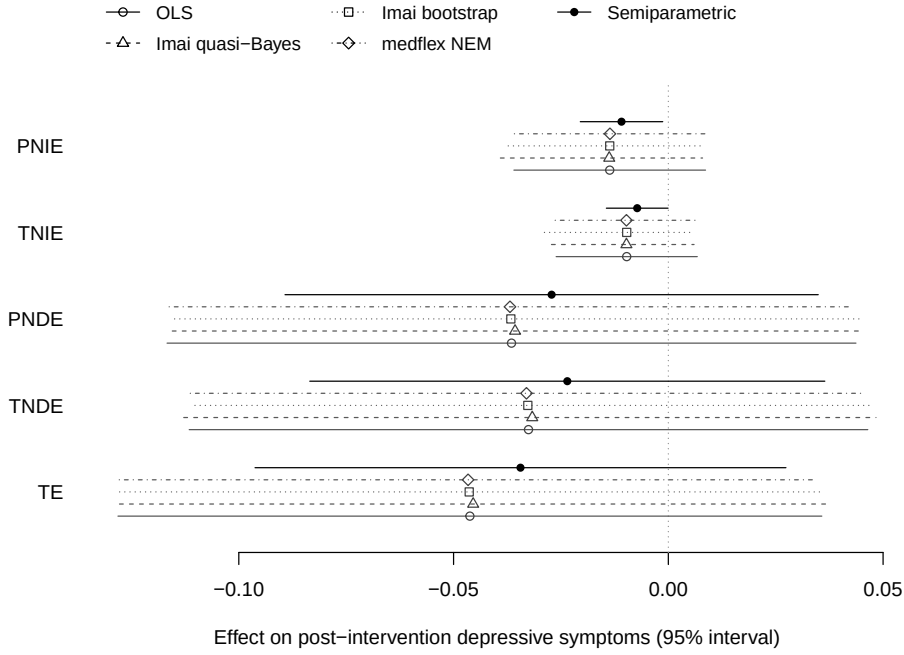


Fig. 4 JOBS II natural-effect estimates for post-intervention depressive symptoms. Points denote estimates, horizontal bars nominal 95% intervals, and the vertical line zero. Negative values indicate lower depressive symptoms under assignment to the intervention

(Table S14). Removing pretreatment depression from the adjustment set changed the proposed PNIE only from -0.0109 to -0.0122 ; both nominal intervals excluded zero (Table S15). This stability check supports the numerical pattern while retaining the baseline-depression-adjusted specification as primary.

Model checks guide the working-model interpretation. OLS residual skewness is -0.781 for the mediator and 1.083 for the outcome, providing substantial non-Gaussian shape information. Full-design residual-variance diagnostics give $p = 0.0228$ and $p = 7.67 \times 10^{-5}$, respectively (Koenker, 1981). Full diagnostics are in Table S16 and Figure S5. We therefore interpret the reported intervals jointly with both the non-Gaussian residual shapes and evidence of variance heterogeneity. The subsequent sensitivity analysis addresses unmeasured mediator–outcome

564 confounding as a separate dimension of the illustration.

565 5.1 Confounding-sensitivity analysis for JOBS II

566 Using the same participants and baseline variables, we fit the reduced-form extension over
 567 $\rho \in [-0.8, 0.8]$. The three-regression stack and its complete nuisance covariance provide a
 568 self-contained calculation of the sensitivity functionals and their uncertainty. Its finite-sample
 569 $\rho = 0$ results complement the primary conditional-outcome estimates in Table 3 as a distinct
 570 reduced-form analysis.

Table 4 JOBS II reduced-form sensitivity analysis. Nominal intervals are pointwise working-model intervals. The zero-effect correlation is conditional on a nonzero mediator treatment effect.

Effect	Method	Effect at $\rho = 0$ (95% CI)	$\hat{\rho}^\dagger$ (95% CI)	$(\hat{\rho}^\dagger)^2$
PNIE	OLS-RF	-0.012 [-0.032, 0.008]	-0.231 [-0.334, -0.122]	0.053
TNIE	OLS-RF	-0.010 [-0.028, 0.007]	-0.213 [-0.292, -0.130]	0.045
PNIE	DS-RF	-0.015 [-0.028, -0.002]	-0.244 [-0.345, -0.137]	0.059
TNIE	DS-RF	-0.013 [-0.025, -0.002]	-0.222 [-0.300, -0.142]	0.049

571 We use the additional variance-decomposition interpretation of Imai et al. (2010a). Let
 572 R_M^{2*} and R_Y^{2*} denote the residual variance fractions explained by a hypothetical pretreatment
 573 confounder. The calibration is $\rho = s\sqrt{R_M^{2*}R_Y^{2*}}$, where $s \in \{-1, 1\}$ specifies opposite or matching
 574 confounder-effect signs. Figure 5 shows both signs. The axes quantify residual-variance fractions
 575 and are distinct from the total-variance quantities $\tilde{R}_M^2, \tilde{R}_Y^2$. This working calibration translates
 576 specified latent variance contributions into sensitivity values; the one-dimensional PNIE/TNIE
 577 curves appear in Figure S6.

578 At $\rho = 0$, OLS-RF gives PNIE -0.0116 with nominal interval $[-0.0316, 0.0084]$, whereas
 579 DS-RF gives -0.0149 with $[-0.0284, -0.0015]$. The corresponding DS-RF TNIE is -0.0132
 580 with $[-0.0246, -0.0018]$. The estimated correlations at which the point estimates cross zero are
 581 -0.231 for OLS-RF and -0.244 for DS-RF for PNIE, corresponding to residual- R^2 products of
 582 0.053 and 0.059 ; TNIE gives -0.213 and -0.222 . Thus opposite-sign confounder associations
 583 explaining roughly 5–6% of the two residual variances in product provide a useful calibration
 584 for the estimated zero boundary. The close boundary estimates agree with their shared first-
 585 order residual-moment representation, while the shorter DS-RF intervals reflect the regression-
 586 coefficient contribution to curve estimation. Reduced-form variance diagnostics give reference
 587 p -values of 0.024 for the pooled mediator model and 0.009 and 0.006 for the two DS-RF outcome

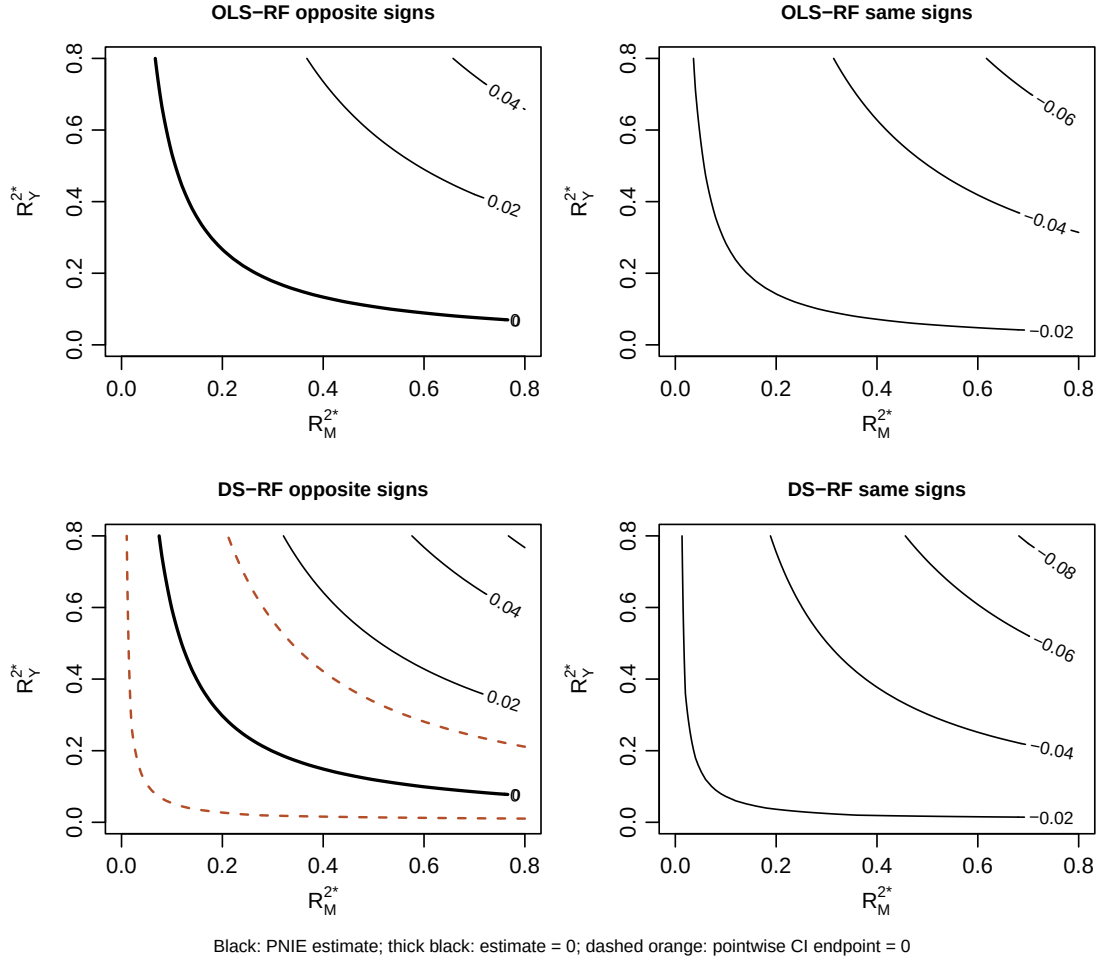


Fig. 5 JOBS II PNIE sensitivity contours from OLS-RF and DS-RF. Each black curve gives combinations of hypothetical residual-variance fractions yielding the labeled point estimate. The thick zero contour concerns the point estimate when crossed in the displayed grid; dashed orange curves mark a pointwise nominal interval endpoint equal to zero. Both confounding signs are shown. The contours are interpreted under the reduced-form working model and alongside the residual-variance diagnostics.

strata. The sensitivity curves are therefore interpreted as pointwise working-model summaries alongside the variance diagnostics and causal assumptions.

6 Discussion

The framework connects error-density information to natural-effect inference and confounding-sensitivity analysis. For regular natural effects under the independent-error mediation model and stated regularity conditions, the joint influence-function representation links regression-level information gains to lower asymptotic variance and higher local power relative to OLS. Simulations support these gains in several non-Gaussian settings, with Gaussian errors providing the OLS-efficient reference.

The independently calibrated experiment connects precision to detection of a weak indirect effect: the advantage persists after accounting for OLS conservatism and against interaction-adapted Huber estimators in the asymmetric-mixture design. Conditional power and all-attempt detection distinguish statistical gains from the probability of returning a usable result.

For the sensitivity functional of Imai et al. (2010a), density-score nuisance estimation and joint covariance propagation improve curve precision in regions identified by the effect gradient. The shared residual-moment representation explains why these gains coexist with nearly unchanged point-estimate zero boundaries. Efficient marginal scores and the assembled estimator's joint influence function support this distinction between curve estimation and confounding sensitivity.

Variance misspecification and unmeasured confounding are separate diagnostic questions. In the asymmetric-mixture experiment, treatment-dependent scaling has the larger impact on calibration. The confounding extension instead relaxes mediator–outcome ignorability while retaining its reduced-form error restrictions. JOBS II illustrates why both dimensions should accompany the effect estimates and nominal intervals.

The deterministic multi-start algorithm and retained fit diagnostics make numerical reliability part of the reproducible evaluation. Covariate-dependent error distributions, incomplete outcomes and comparisons with flexible mediation estimators are directions for further development. The present framework provides a theoretical and computational basis for using error-shape information in mediation inference under explicit assumptions.

Statements and Declarations

Competing interests. The author declares no financial or non-financial competing interests.

Funding. No external funding was received for this study.

Data availability. The JOBS II illustration data are distributed with version 4.5.1 of the R package **mediation**. The exact data object used for this analysis is included in the project repository identified below.

Ethics approval and consent to participate. This methodological study uses publicly distributed, de-identified JOBS II illustration data. It involved no new participant recruitment or participant identification; the original intervention is described by Vinokur and Schul (1997).

Code availability. The code and reproducibility materials are publicly available at <https://github.com/mijeong-kim/semi-causal-med>. The repository contains the analysis scripts and seeds, the JOBS II data object, retained numerical results, tables, figures, package versions and session information. Its **README.md** provides the software requirements and commands for reproducing the analyses and rebuilding the tables and figures from the retained results.

Supplementary information. Online Resource 1 (Supplementary Material) contains proofs, implementation details, complete numerical results and diagnostics. Code and reproducibility materials are provided through the GitHub repository listed above.

Author contributions. Mijeong Kim developed the statistical methodology, conducted the analyses, and wrote the manuscript.

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